



Document Management System



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Title: Chest Drain Insertion

1.0 Indications

- Treatment of pneumothorax or pleural effusion

2.0 Equipment

- Ø Sterile dressing pack
- Ø Cleaning solution as per unit policy – suggestaqueous chlorhexidine 2%/isopropyl alcohol70% (Chloraprep) and wash off withsodium chloride 0.9% once dried for babies <26 weeks' gestation
- Ø Lidocaine 1%, with syringe and needle forpreparation and injection
- Ø Chest drains size FG 8,10,12 (use largest possibledepending on size of baby)
- Ø Low pressure suction unit
- Ø Scalpel and fine straight blade (size 11)
- Ø Fine blunt forceps
- Ø Underwater seal chest drainage bottle and tubing
- Ø Steristrips and transparent dressing(e.g.Opsite/Tegaderm)

3.0 Sites

- Ø *Site of insertiondepends on position of pneumothorax
 - preferred site is in anterior axillary line, betweenfourth and sixth intercostal space, to concealsubsequent scarring and avoidinterference with breast development
 - alternative site is just lateral to midclavicularline, in second or third intercostal space

- for pleural effusion, use midaxillary line between fourth and fifth intercostal spaces, and direct drain posteriorly

4.0 Procedure

4.1 Preparation and position of baby

- ü Inform parents and obtain verbal consent (unless emergency procedure)
- ü Use 10–12 FG pleural catheter (small babies may need 8 FG)
- ü Position baby supine and flat with affected side slightly tilted up (for example, by using a folded blanket)
- ü Prepare skin with full aseptic technique
- ü Infiltrate with lidocaine 1%, **even in babies being given systemic analgesia**

4.2 Insertion of tube

- o * Make small incision in skin with scalpel at lower edge of intercostal space to avoid injury to intercostal vessels
- o * Dissect bluntly with fine forceps through intercostal muscle and pleura
- o * Use fine forceps to gently advance tip of catheter
- o * Push and twist tube gently through incision into pleural space
- o * Insert chest drain 2–3 cm for small preterm and 3 cm for term babies
- o * Use of trocar not generally recommended. If used (in bigger baby), protect lung by clamping artery forceps onto trocar 1 cm from the tip
- o * Connect tube to prepared underwater seal
- o * Manipulate tube gently so that tip lies anteriorly in thoracic cavity for pneumothorax, and posteriorly for effusion
- o * Secure tube with SteriStrips, and cover with gauze dressing. A suture may be required; **do not use purse-string suture**
- o * Secure tube to chest wall using suitable tape (Opsite or Tegaderm)

5.0 Aftercare

- Ø * Check bubbling or oscillation of water column seen with every inspiration
- Ø * Check tube position with chest X-ray (consider lateral x-ray to confirm position)

5.1 Suction

- Ø If bubbling poor and X-ray confirms drain is in incorrect position but pneumothorax not fully draining on X-ray or cold light, apply continuous suction of 5–10 cm of water. Occasionally, a second drain may be necessary

5.2 Document

- Ø * Record presence of bubbling (continuous/intermittent/none) in nursing care chart
- Ø * Record with nursing observations, bubbling and/or oscillation of water column seen with every inspiration

6.0 Removal of chest drain

- o Remove tubing when no bubbling or oscillation of water column has occurred for 24 hr.

- o After removing drain, close wound with steristrips,a suture is seldom necessary
- o Close clinical observation after removal of drain issufficient to diagnose re-accumulation ofthe air leak, routine chest X-ray notgenerally warranted

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Neonatal Guidelines 2015–17, by The Bedside Clinical Guidelines Partnership	NHS	2015	47

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